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(54) **LOCATION BASED ACCESS CONTROL FOR CONTENT DELIVERY NETWORK RESOURCES**

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Related U.S. Application Data

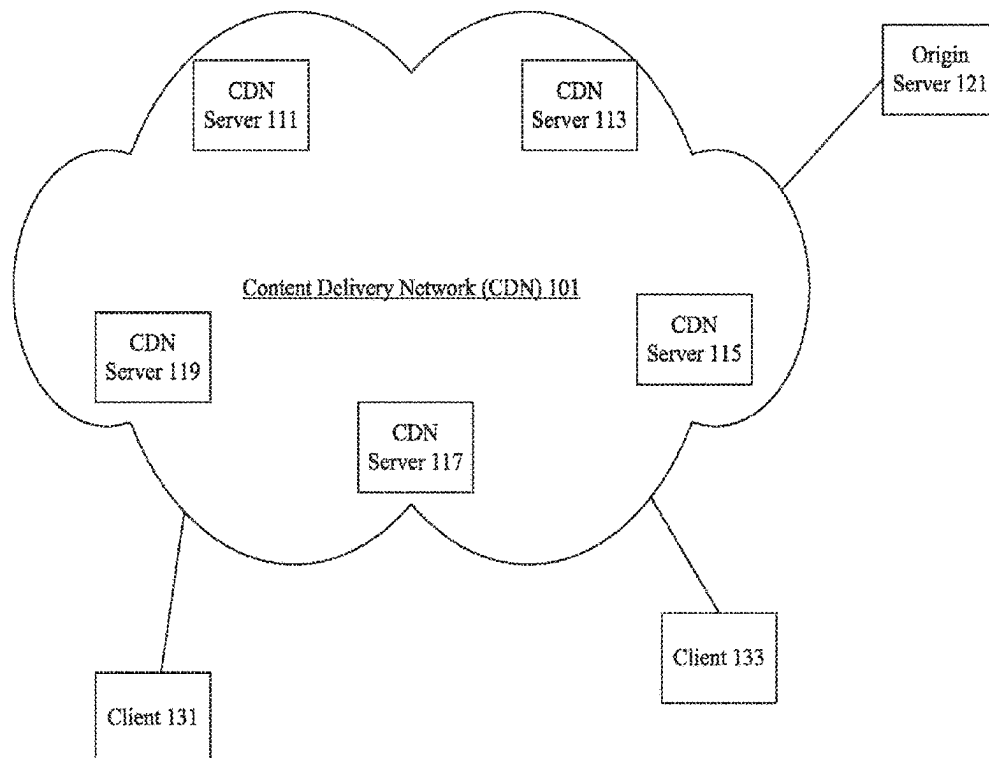
(63) Continuation of application No. 18/743,693, filed on Jun. 14, 2024, now Pat. No. 12,309,220, which is a continuation of application No. 17/688,468, filed on Mar. 7, 2022, now Pat. No. 12,041,146, which is a continuation of application No. 16/555,999, filed on Aug. 29, 2019, now Pat. No. 11,303,716, which is a continuation of application No. 15/188,951, filed on Jun. 21, 2016, now Pat. No. 10,447,801, which is a continuation of application No. 14/495,815, filed on Sep. 24, 2014, now Pat. No. 9,398,112, which is a continuation of application No. 13/074,989, filed on Mar. 29, 2011, now Pat. No. 8,874,750.

Publication Classification

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(57) **ABSTRACT**

Mechanisms are provided to allow efficient delivery of content through a content delivery network (CDN) without taxing an origin server while maintaining fine grained location based access control. Requests to a CDN are augmented with quantized and non-quantized versions of location information. A CDN can use quantized location information in a URI to identify and return content if content corresponding to the URI is available in cache. If the content is not available in cache, the CDN can contact an origin server to obtain a result for the request using non-quantized location information. The origin server examines the non-quantized location information to determine whether to respond to the client request with content. The origin server examines the area corresponding to the quantized location information to determine whether the content should be cached at the CDN.



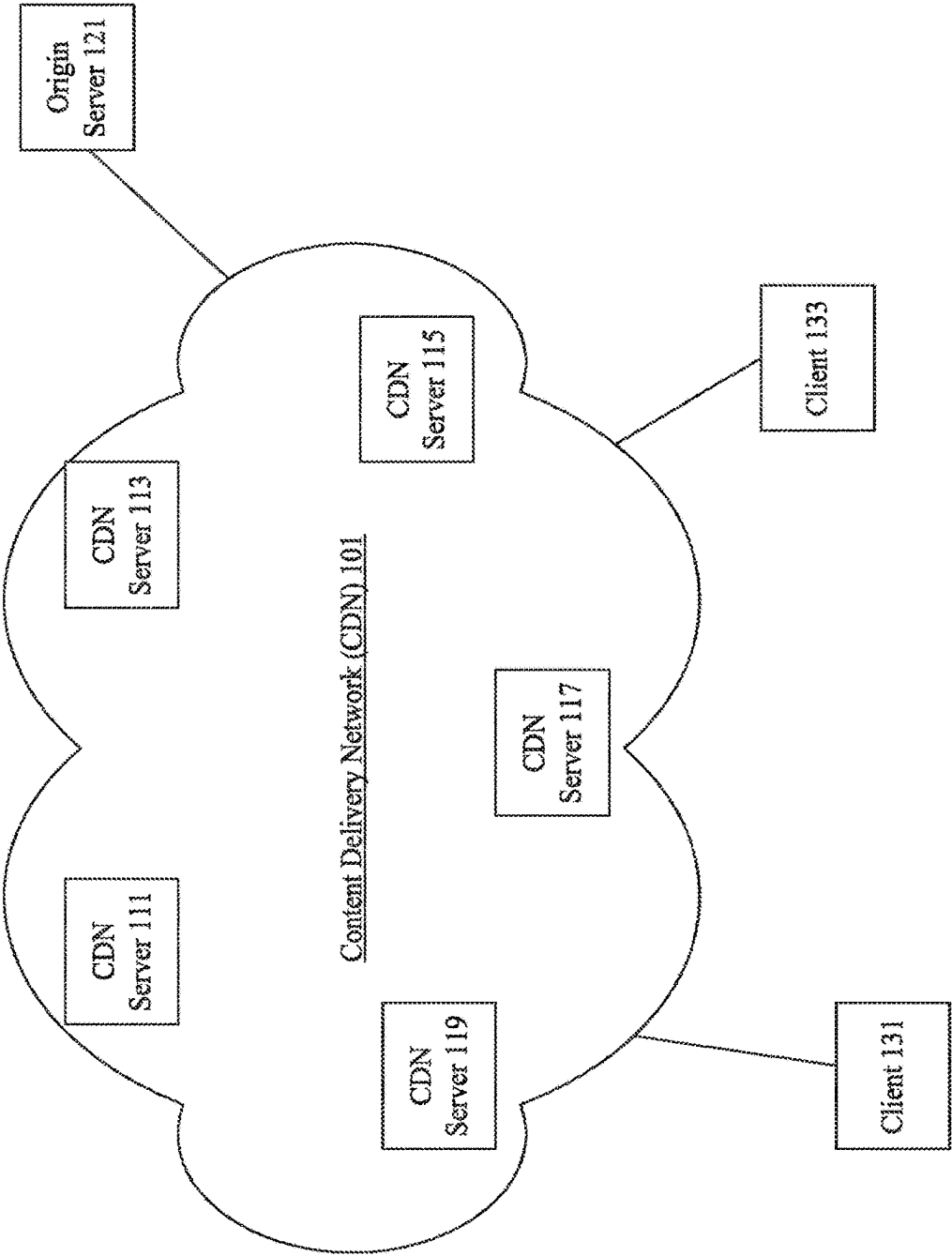


Figure 1

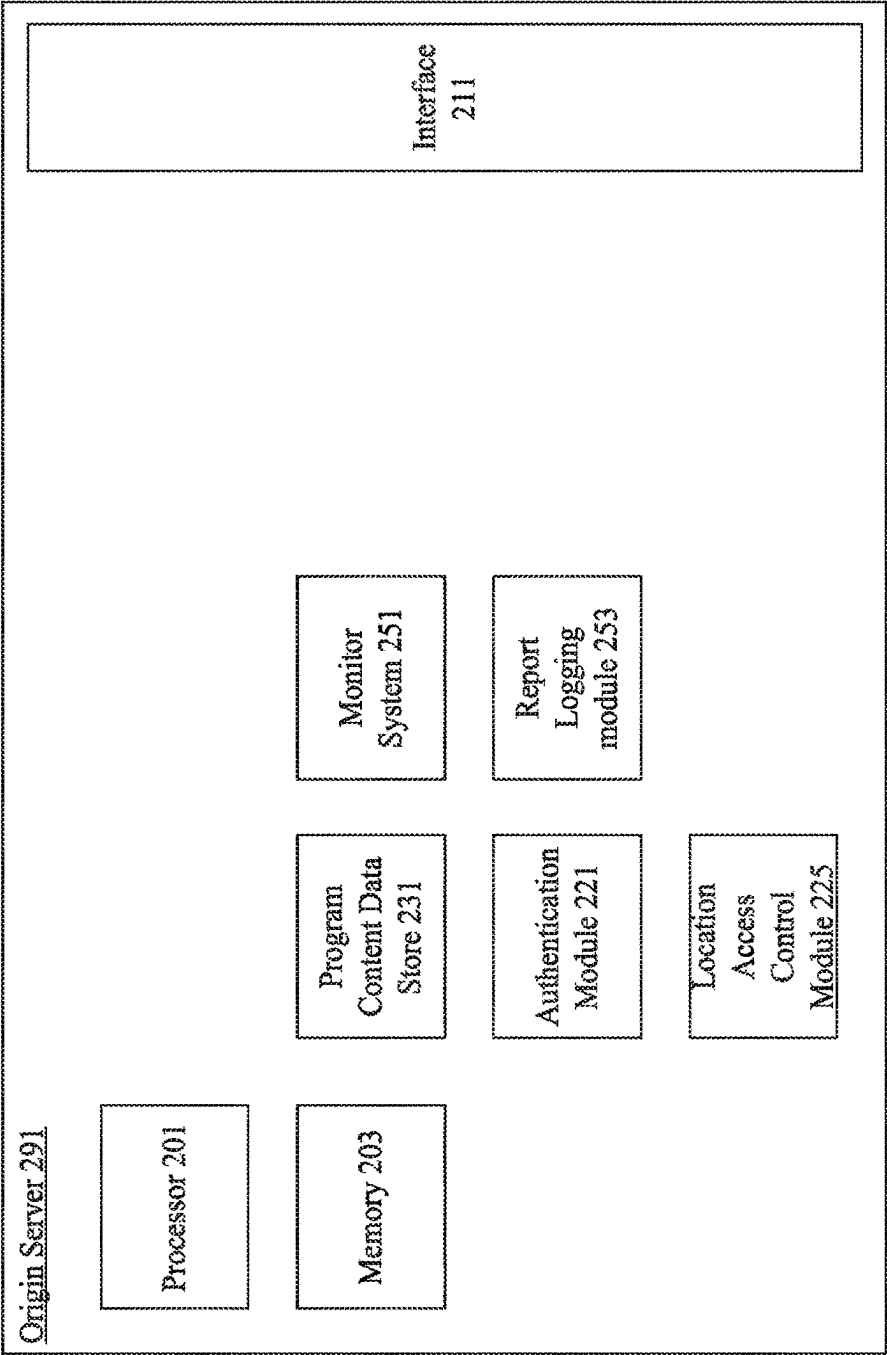


Figure 2

Location Restricted Resource Identifier 301	Quantized Location Coordinates 303	Non-Quantized Location Coordinates 305
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Figure
3

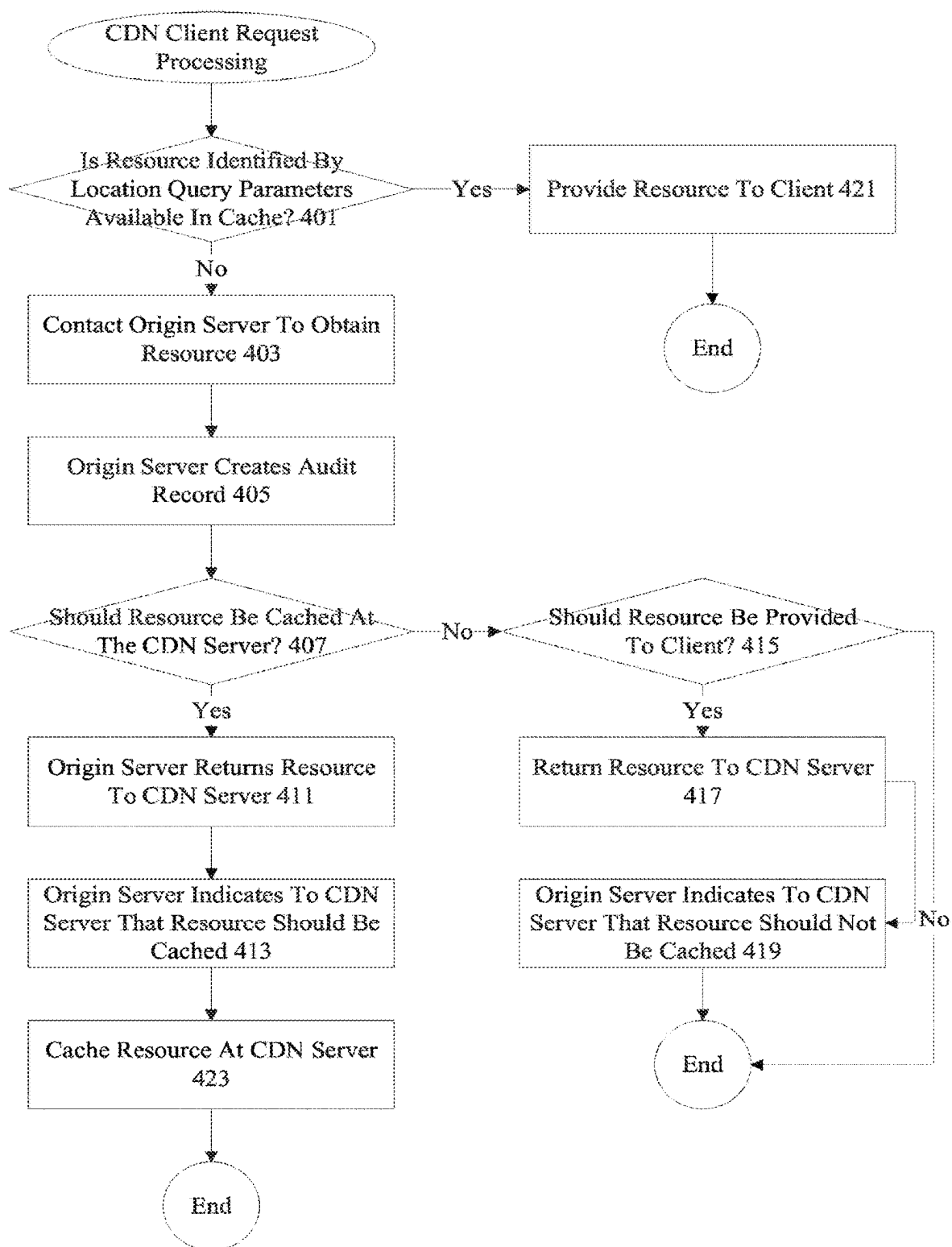


Figure
4

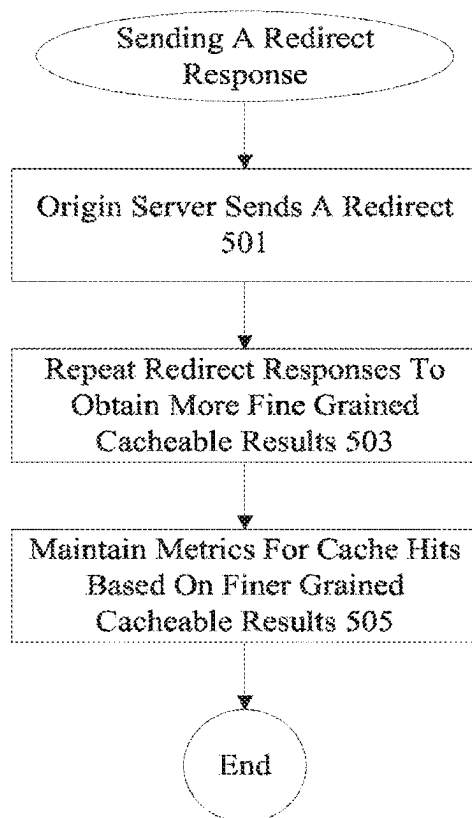


Figure
5

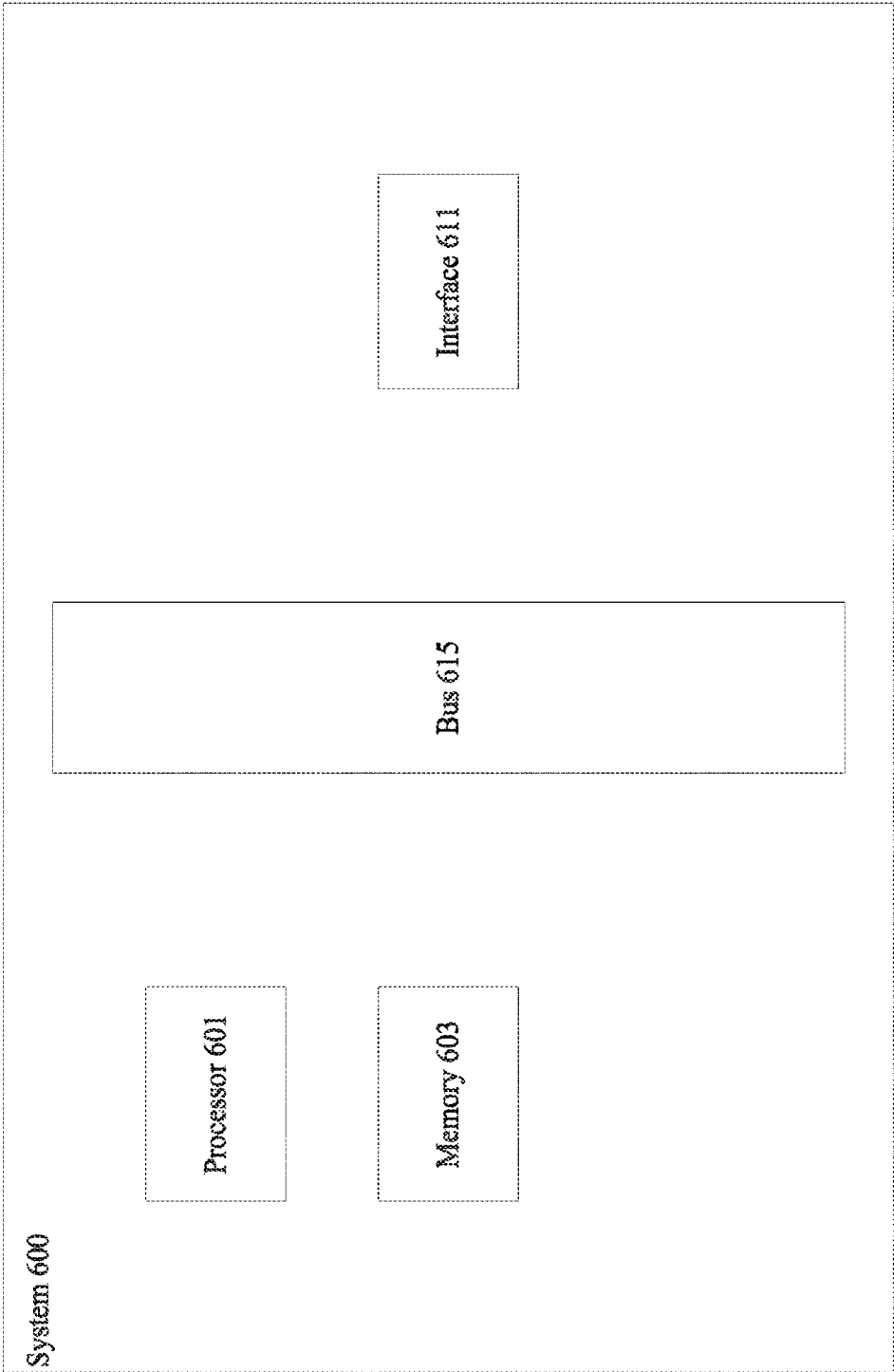


Figure 6

LOCATION BASED ACCESS CONTROL FOR CONTENT DELIVERY NETWORK RESOURCES

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is a continuation of and claims priority to U.S. patent application Ser. No. 15/188,951 (Atty Docket No. MOBIP063C2), filed Jun. 21, 2016, by James Roseborough and Stephen Coney, which is a continuation of and claims priority to U.S. patent application Ser. No. 14/495,815 (Atty Docket No. MOBIP063C1), filed Sep. 24, 2014, by James Roseborough and Stephen Coney, which is a continuation of and claims priority to U.S. patent application Ser. No. 13/074,989 (Atty Docket No. MOBIP063US), now U.S. Pat. No. 8,874,750, filed Mar. 29, 2011, by James Roseborough and Stephen Coney which is hereby incorporated by reference in its entirety and for all purposes.

TECHNICAL FIELD

[0002] The present disclosure relates to a mechanism for providing location based access control for content delivery network resources.

DESCRIPTION OF RELATED ART

[0003] It is often desirable to use content delivery networks (CDNs) to distribute resources such as media content to clients. CDNs have scalable network and server capacity to meet client demand. However, CDNs do not typically allow for fine grained access control to resources. Consequently, the techniques and mechanisms of the present invention provide improved mechanisms for providing location based access control.

OVERVIEW

[0004] Mechanisms are provided to allow efficient delivery of content through a content delivery network (CDN) without taxing an origin server while maintaining fine grained location based access control. Requests to a CDN are augmented with quantized and non-quantized versions of location information. A CDN can use quantized location information in a URI to identify and return content if content corresponding to the URI is available in cache. If the content is not available in cache, the CDN can contact an origin server to obtain a result for the request using non-quantized location information. The origin server examines the non-quantized location information to determine whether to respond to the client request with content. The origin server examines the area corresponding to the quantized location information to determine whether the content should be cached at the CDN.

[0005] These and other features of the present invention will be presented in more detail in the following specification of the invention and the accompanying figures, which illustrate by way of example the principles of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] The disclosure may best be understood by reference to the following description taken in conjunction with the accompanying drawings, which illustrate particular embodiments of the present invention.

[0007] FIG. 1 illustrates a particular example of a network that can use the techniques and mechanisms of the present invention.

[0008] FIG. 2 illustrates a particular example of an origin server.

[0009] FIG. 3 illustrates a particular example of a client request.

[0010] FIG. 4 illustrates a particular example of a content delivery network (CDN) processing a client request.

[0011] FIG. 5 illustrates a particular example of redirect request.

[0012] FIG. 6 illustrates a particular example of a client device.

DESCRIPTION OF PARTICULAR EMBODIMENTS

[0013] Reference will now be made in detail to some specific examples of the invention including the best modes contemplated by the inventors for carrying out the invention. Examples of these specific embodiments are illustrated in the accompanying drawings. While the invention is described in conjunction with these specific embodiments, it will be understood that it is not intended to limit the invention to the described embodiments. On the contrary, it is intended to cover alternatives, modifications, and equivalents as may be included within the spirit and scope of the invention as defined by the appended claims.

[0014] For example, the techniques of the present invention will be described in the context of particular devices such as mobile devices. However, it should be noted that the techniques and mechanisms of the present invention can be used with a variety of devices including general computing devices. In the following description, numerous specific details are set forth in order to provide a thorough understanding of the present invention. The present invention may be practiced without some or all of these specific details. In other instances, well known process operations have not been described in detail in order not to unnecessarily obscure the present invention.

[0015] Various techniques and mechanisms of the present invention will sometimes be described in singular form for clarity. However, it should be noted that some embodiments include multiple iterations of a technique or multiple instantiations of a mechanism unless noted otherwise. For example, a processor is used in a variety of contexts. However, it will be appreciated that multiple processors can also be used while remaining within the scope of the present invention unless otherwise noted. Furthermore, the techniques and mechanisms of the present invention will sometimes describe two entities as being connected. It should be noted that a connection between two entities does not necessarily mean a direct, unimpeded connection, as a variety of other entities may reside between the two entities. For example, a processor may be connected to memory, but it will be appreciated that a variety of bridges and controllers may reside between the processor and memory. Consequently, a connection does not necessarily mean a direct, unimpeded connection unless otherwise noted.

[0016] Many organizations rely on content delivery networks (CDNs) to efficiently deliver content to clients. CDNs generally have the network bandwidth and server capacity to scale up and down with client demand. It is often more economical for a content publisher to partner with a CDN to

deliver content. Otherwise, the publisher may end up under-investing or overinvesting in hardware and network capacity.

[0017] CDNs typically function by receiving requests from clients, checking a local cache for a copy of the requested resource, querying the origin server belonging to the publisher if the requested resource is missing from the cache or has expired, and then returning the resource to the client and storing it in cache for use with future requests. CDN servers are typically strategically located at the edges of various networks to limit loads on network interconnects and backbones. CDN servers are often redundantly deployed and interact with other CDN servers to respond to content requests from clients and optimize content delivery. Optimization may include bandwidth usage reduction, latency reduction, and improved availability.

[0018] However, a shortcoming of CDNs is that they typically do not allow for fine-grained access-control to network resources. Once the content is in cache, the function of access-control is left to the CDN. The access control systems employed by CDNs usually are meant to serve the greatest common denominator. This does not permit the fine-grained access-control that some organizations require.

[0019] Consequently, the techniques and mechanisms of the present invention allow location based or geographically allocated resources to be cached within a CDN while still permitting a fine-grained proprietary location based access-control algorithm to be applied by the origin server or content publisher. According to various embodiments, the CDN does not possess any knowledge or proprietary algorithm details, or that a proprietary algorithm even exists.

[0020] Some CDNs support location and geolocation checks common to all customers of the CDN. The CDN will typically prevent access to resources unless they pass certain geolocation checks. These checks are often based on source internet protocol (IP) address, which in a mobile environment is often unreliable. Other CDNs will rely entirely on an origin server for fine-grained access-control to resources. Requests received at a CDN server may be routed to an origin server for authorization. However, routing numerous requests to an origin server eliminates many of the benefits of using a CDN in the first place.

[0021] According to various embodiments, a system for providing fine grained location based access control includes an origin server, a CDN server, and a client. A client may be a mobile device, smartphone, computing system, etc. According to various embodiments, a client obtains location from using services such as global positioning system (GPS), cell tower triangulation, wireless fidelity (WiFi), radio frequency identification (RFID), near field communication (NFC), etc. In particular embodiments, GPS longitude and latitude coordinates are obtained using device based location application program interfaces (APIs). Using location information, the client sends a request for content that gets routed to a CDN. In particular embodiments, the request is based on a quantized version of the latitude and longitude as reported by the device based location APIs. For instance if the latitude and longitude is 37.75098874, 122.47389284, the quantized latitude and longitude to one significant decimal point would be 37.8, 122.5.

[0022] According to various embodiments, the client request to the CDN would include the quantized latitude and longitude in a cacheable portion of the uniform resource locator (URL) or as query parameters of the URL. It is

recognized that the degree of quantization may vary. Queries with particular quantized latitude and longitude values will correspond to particular resources cached at the CDN. In particular embodiments, the client request to the CDN would include the non-quantized latitude and longitude in a non-cacheable portion of the request such as headers or the query parameters of the URL. According to various embodiments, the CDN determines whether the content or resource is currently in cache. For example, the CDN would determine whether the content is currently in cache by using a query that includes quantized location information 37.8 and 122.5. If the resource such as media content is not cached, the CDN contacts the origin server to obtain the result of the request. The determination of whether a client is authorized to access location restricted content can be based on the more accurate non quantized latitude and longitude parameters at the origin server, while the determination is based on quantized latitude and longitude parameters at the CDN server. The origin server may create an audit record indicating the requested resource, the time of request and the returned result. The result returned will indicate whether the CDN should cache the content.

[0023] According to various embodiments, the origin server determines whether the CDN should cache the content based on whether the region defined by the quantized location information (e.g. 37.8 and 122.5) should have access to a particular piece of content. In some examples, the region defined by 37.8 and 122.5 should have access and the origin server instructs the CDN to cache the content to allow the CDN to efficiently respond to queries with the quantized location coordinates 37.8 and 122.5. In other examples, only part of the region defined by 37.8 and 122.5 should have access to the content. The 37.8 and 122.5 area may be the boundary region of a location based blackout zone. Consequently, the origin server directs the CDN not to cache the content. Requests from the 37.8 and 122.5 areas are forwarded to the origin server which handles responding to the requests using more fine grained non-quantized longitude and latitude values.

[0024] If the content or resource corresponding to a query with quantized location information is already cached, the content or resource is supplied from cache. The origin server may send a **302** re-direct with a quantized result with more accuracy or significant digits based on the non-quantized supplied latitude and longitude parameters. Such **302** re-direct responses are optional and could be repeated to obtain more fine grained cacheable results for boundary conditions of location based exclusion zones. According to various embodiments, content can be delivered quickly and efficiently through a CDN without taxing the origin server and without losing fine grained location based access control. This allows the CDN to excel in providing high capacity, high reliability, and low latency while allowing the publisher to maintain strict control over the access to location restricted content.

[0025] FIG. 1 illustrates one example of a CDN that can be used with various embodiments. According to various embodiments, a CDN **101** includes CDN servers **111**, **113**, **115**, **117**, and **119**. In particular embodiments, CDN servers are strategically deployed to enhance content delivery performance. CDN servers may also be referred to surrogate servers or content replica servers. Effective CDN server placement may reduce the number of servers needed as well as the number of times content has to be replicated. A variety

of algorithms can be used to deploy CDNs. Greedy algorithms continually make locally optimal choices with the hope of finding a global optimum. Hot spot algorithms place CDN servers near the clients generating the greatest load. Treebased algorithms specify the locations of CDN servers to achieve particular levels of performance.

[0026] The number of CDN servers may vary from the dozens to thousands and distribute content from origin server **121**. Although an origin server **121** may have very specific mechanisms for determining what clients have access to particular pieces of content, once the content is distributed onto a CDN **101**, the origin server **121** has limited access control mechanisms. For example, client **131** may be authorized to access particular geographically restricted piece of content based on the location of the client **131** but a client **133** in a different geographic area may not. Alternatively, different versions of content may be provided to different geographic regions.

[0027] According to various embodiments, a client request for content is typically algorithmically directed at a CDN server that can efficiently server the client request. In order to verify that a particular client **131** has access to a piece of geographically restricted content, a request may be sent to an origin server **121** to perform location verification. However, location verification is typically based on internet protocol (IP) address, which is not a very accurate location indicator, particularly for mobile devices.

[0028] FIG. 2 is a diagrammatic representation showing one example of an origin server **291**. According to various embodiments, the origin server **291** includes a processor **201**, memory **203**, and an interface. In some examples, the interfaces include a program content interfaces, buy engine interfaces, carrier interfaces, etc., to allow the origin server **291** the ability to interface with external entities including program content providers, carriers, vendors, billing systems, purchase engines, etc. The origin server **291** also can include a program content data store **231** configured to store program content such as video clips, pay per view content, movies, programs, and live or near-live streams. The origin server **291** can also maintain static information such as icons and menu pages.

[0029] An authentication module **221** verifies the identity of mobile devices. Location access control module **225** associated with authentication module **221** determines a client location to determine whether the client has access to a particular piece of content. For example, a location access control module may determine that a client while located in a geographic market X should not have access to a sports program. However, if the client moves to a location in a geographic market Y, the client should have access to the same sports program. Alternatively, the location access control module may determine that a client in a geographic market X should have access to a different news cast than a client in a geographic market Y even if the clients request the same piece of news programming.

[0030] In many implementations without CDNs, the origin server **291** can determine the location of the client using location information associated with the client. However, when CDNs distribute the content, CDNs typically do not allow for the same degree of access control and may provide a client with requested content regardless of geographic location. Alternatively, the CDN may forward client requests for content to an origin server **291** for the location access control module **225** to handle location restrictions. However,

forwarding requests to the origin server **291** removes some of the primary benefits of using CDNs.

[0031] A logging and report generation module **253** tracks mobile device requests and associated responses. A monitor system **251** allows an administrator to view usage patterns and system availability. According to various embodiments, the origin server **291** handles requests and responses for media content related transactions and provides actual content. In particular embodiments, requests for content and actual content distribution can be handled by separate servers. In some embodiments, the origin server **291** can also be configured to provide media clips and files to a client in a manner that supplements a streaming server.

[0032] Although a particular origin server **291** is described, it should be recognized that a variety of alternative configurations are possible. For example, some modules such as a report and logging module **253** and a monitor **251** may not be needed on every server. Alternatively, the modules may be implemented on another device connected to the server. In another example, the server **291** may include a concrete buy engine implementation. A variety of configurations are possible.

[0033] FIG. 3 illustrates a particular example of a client request. According to various embodiments, a client request to a CDN includes a location restricted content identifier **301**, quantized latitude and longitude information in a cacheable portion **303** of the uniform resource identifier (URI), as well as non-quantized latitude and longitude information in a non-cacheable portion **305**. The cacheable portion may be the path of the URI or query parameters of a uniform resource locator (URL). The non-cacheable portion may be a header or other query parameters of the URL. According to various embodiments, the CDN may have content in cache to respond to a request including quantized longitude and latitude parameters. If the CDN has content to respond to the request, the content is provided to the client. If the CDN does not have content to respond to the request, the request is forwarded to an origin server to determine whether the client should have access to location restricted content.

[0034] FIG. 4 illustrates a particular example of a CDN processing a client request. According to various embodiments, the CDN determines whether a resource is currently in cache at **401**. If the resource is not currently in cache, the CDN contacts the origin server at **403** to obtain the result of the request. According to various embodiments, the determination is based on the more accurate non-quantized latitude and longitude parameters. At **405**, the origin server creates an audit record identifying the requested resource, the time of the request, and the returned result. At **407**, the origin server determines whether the content should be cached at a CDN. According to various embodiments, the origin server determines whether the geographic area defined by the quantized location value should have access to location restricted content. In particular embodiments, the area defined by 39.5 and 122.8 may fall entirely within a region that is allowed access to a particular sports program. The origin server returns the content to the CDN server at **411** and indicates to the CDN that the content should be cached **413**. The resource is then cached at **415**. The content can then subsequently be provided to any subscriber sending a request with the quantized location parameters 39.5 and 122.8.

[0035] In other embodiments, only part of the area defined by 39.5 and 122.8 should be allowed access to a particular piece of content 415. According to various embodiments, the area may cover a boundary line that separates a blackout zone from a non-blackout zone. Consequently, the origin server makes a determination of whether to send the content to the client based on more precise non-quantized location parameters, returns the resource to the CDN at 417, and indicates to the CDN that the content should not be cached at 419. If the resource is cached at 401, the result is supplied from cache at 421.

[0036] FIG. 5 illustrates one particular embodiment for sending redirect responses to provide for access control with finer granularity. According to various embodiments, an origin server sends a redirect such as a 302 re-direct at 501. The redirect has a quantized result with more accuracy or significant digits based on the non-quantized supplied latitude and longitude parameters. At 503, redirect responses are repeated to obtain more fine grained cacheable results for boundary conditions of location based exclusion zones. Location based blackouts for location restricted media content can be implemented with precision. In particular embodiments, different content can be provided to different clients based on quantized location information. At 505, metrics are maintained for cache hits based on finer grained cacheable results. The level of granularity can be adjusted to provide for additional granularity if there are excessive requests forwarded to an origin server.

[0037] FIG. 6 illustrates one example of a server that can be used to provide location based access control. According to particular embodiments, a system 600 suitable for implementing particular embodiments of the present invention includes a processor 601, a memory 603, an interface 611, and a bus 615 (e.g., a PCI bus or other interconnection fabric) and operates as a streaming server. When acting under the control of appropriate software or firmware, the processor 601 is responsible for modifying and transmitting live media data to a client. Various specially configured devices can also be used in place of a processor 601 or in addition to processor 601. The interface 611 is typically configured to end and receive data packets or data segments over a network.

[0038] Particular examples of interfaces supports include Ethernet interfaces, frame relay interfaces, cable interfaces, DSL interfaces, token ring interfaces, and the like. In addition, various very high-speed interfaces may be provided such as fast Ethernet interfaces, Gigabit Ethernet interfaces, ATM interfaces, HSSI interfaces, POS interfaces, FDDI interfaces and the like. Generally, these interfaces may include ports appropriate for communication with the appropriate media. In some cases, they may also include an independent processor and, in some instances, volatile RAM. The independent processors may control such communications intensive tasks as packet switching, media control and management.

[0039] According to various embodiments, the system 600 is a content server that also includes a transceiver, streaming buffers, and a program content database. The content server may also be associated with subscription management, logging and report generation, and monitoring capabilities. In particular embodiments, functionality for allowing operation with mobile devices such as cellular phones operating in a particular cellular network and providing subscription management. According to various embodiments, an

authentication module verifies the identity of devices including mobile devices. A logging and report generation module tracks mobile device requests and associated responses. A monitor system allows an administrator to view usage patterns and system availability. According to various embodiments, the content server 691 handles requests and responses for media content related transactions while a separate streaming server provides the actual media streams. [0040] Because such information and program instructions may be employed to implement the systems/methods described herein, the present invention relates to tangible, machine readable media that include program instructions, state information, etc. for performing various operations described herein. Examples of machine-readable media include hard disks, floppy disks, magnetic tape, optical media such as CD-ROM disks and DVDs; magneto-optical media such as optical disks, and hardware devices that are specially configured to store and perform program instructions, such as read-only memory devices (ROM) and programmable read-only memory devices (PROMs). Examples of program instructions include both machine code, such as produced by a compiler, and files containing higher level code that may be executed by the computer using an interpreter.

[0041] While the invention has been particularly shown and described with reference to specific embodiments thereof, it will be understood by those skilled in the art that changes in the form and details of the disclosed embodiments may be made without departing from the spirit or scope of the invention. It is therefore intended that the invention be interpreted to include all variations and equivalents that fall within the true spirit and scope of the present invention.

1-20. (canceled)

21. A method comprising:

receiving, from a client device, a first request for a first resource of a first type;

based at least in part on the first request being a request for a resource of the first type:

evaluating first location coordinates of the client device provided at a first level of precision; and

providing the first resource to the client device based on the evaluating the first location coordinates of the client device;

receiving, from the client device, a second request for a second resource of a second type;

based at least in part on the second request being a request for a resource of the second type:

evaluating second location coordinates of the client device provided at a second level of precision, wherein second level of precision is more precise than the first level of precision; and

providing the second resource to the client device based on the evaluating the second location coordinates of the client device.

22. The method of claim 21, wherein:

the first type of resource is a resource already in cache; and

the second type of resource is a resource not yet stored in the cache.

23. The method of claim 21, wherein the second level of precision being more precise than the first level of precision is based on the second location coordinates comprising a higher number of digits than the first location coordinates.

24. The method of claim **21**, wherein the second level of precision being more precise than the first level of precision is based on the second location coordinates comprising a higher number of significant figures than the first location coordinates.

25. The method of claim **21**, wherein the first location coordinates are quantized location coordinates and the second location coordinates are non-quantized location coordinates.

26. The method of claim **21**, further comprising:
obtaining the first location coordinates from at least one of
a global positioning system (GPS) device or a device-based location application program interface (API).

27. The method of claim **21**, wherein the providing the second resource to the client device is further based on determining that the client device is authorized to receive the second resource based on subscription information of the client device.

28. The method of claim **21**, wherein the evaluating the second location coordinates comprises:

determining that the second location coordinates are within an area wherein the client device is allowed access to the second resource.

29. The method of claim **21**, further comprising:
sending a request to an origin server for the second location coordinates at a level of precision higher than the second level of precision, wherein the origin server is configured to increase the level of precision based at least in part on receiving successive requests.

30. A system comprising:

an input/output circuitry configured to:

receive, from a client device, a first request for a first resource of a first type;

a control circuitry configured to:

based at least in part on the first request being a request for a resource of the first type:

evaluate first location coordinates of the client device provided at a first level of precision; and

provide the first resource to the client device based on the evaluating the first location coordinates of the client device;

wherein the input/output circuitry is further configured to:

receive, from the client device, a second request for a second resource of a second type; and

wherein the control circuitry is further configured to:

based at least in part on the second request being a request for a resource of the second type:

evaluate second location coordinates of the client device provided at a second level of precision, wherein second level of precision is more precise than the first level of precision; and

provide the second resource to the client device based on the evaluating the second location coordinates of the client device.

31. The system of claim **30**, wherein:

the control circuitry is further configured to:

store the first type of resource in cache; and

store the second type of resource in the cache.

32. The system of claim **30**, wherein the second level of precision being more precise than the first level of precision is based on the second location coordinates comprising a higher number of digits than the first location coordinates.

33. The system of claim **30**, wherein the second level of precision being more precise than the first level of precision is based on the second location coordinates comprising a higher number of significant figures than the first location coordinates.

34. The system of claim **30**, wherein the first location coordinates are quantized location coordinates and the second location coordinates are non-quantized location coordinates.

35. The system of claim **30**, wherein the control circuitry is further configured to obtain the first location coordinates from at least one of a global positioning system (GPS) device or a device-based location application program interface (API).

36. The system of claim **30**, wherein the control circuitry is further configured to:

determine that the client device is authorized to receive the second resource based on subscription information of the client device.

37. The system of claim **30**, wherein the control circuitry is further configured to:

determine that the second location coordinates are within an area wherein the client device is allowed access to the second resource.

38. The system of claim **30**, wherein the control circuitry is further configured to:

send a request to an origin server for the second location coordinates at a level of precision higher than the second level of precision, wherein the origin server is configured to increase the level of precision based at least in part on receiving successive requests.

* * * * *